

Clustering and Indexing

Database Modelling and Design

Nikhil Negi (153)
Anmol (135)
Vishnu (136)

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Introduction

- Databases store large volumes of structured data
- Efficient data retrieval is critical for performance
- Poor data access leads to slow queries
- Two techniques: Clustering and Indexing
- Improves query execution time

What is Clustering?

- Grouping related records together physically
- Based on common attributes
- Reduces disk I/O operations
- Improves query performance

Types of Clustering

- Simple Clustering (single attribute)
- Composite Clustering (multiple attributes)
- Physical Clustering (stored in sorted order)

Advantages of Clustering

- Faster data retrieval
- Reduces disk access
- Efficient for range queries
- Improves join operations

Disadvantages of Clustering

- Complex implementation
- Requires maintenance
- Not suitable for frequent updates
- Higher modification overhead

What is Indexing?

- Technique to quickly locate data
- Similar to index in a book
- Stores key and pointer
- Avoids full table scan

Types of Indexing

- Primary Index (unique key)
- Secondary Index (non-key)
- Clustered Index (physical order)
- Non-Clustered Index (separate structure)

Primary vs Secondary Index

Primary Index

- Based on primary key
- Unique values
- Sorted order

Secondary Index

- Non-primary attribute
- Duplicate values allowed
- Flexible searching

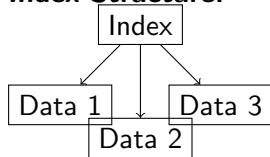
Advantages of Indexing

- Faster query execution
- Reduces search time
- Efficient for large data
- Improves SELECT queries

Clustering vs Indexing + Diagram

Feature	Clustering	Indexing
Purpose	Group data	Locate data
Storage	Physical	Logical
Speed	Range queries	Search queries

Index Structure:



Thank You

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